

# Speech and Audio Processing

## 04 - speech synthesis 2

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# Class schedule (tentative)

|           |                |                                       |                                                               |
|-----------|----------------|---------------------------------------|---------------------------------------------------------------|
| 01        | Oct. 03        | Introduction                          | Learn from basic to state-of-the-art about speech processing. |
| 02        | Oct. 10        | Speech production / 音声生成              |                                                               |
| 03        | Oct. 17        | Speech synthesis / 音声合成               |                                                               |
| <b>04</b> | <b>Oct. 24</b> | <b>Speech synthesis 2 / 音声合成2</b>     |                                                               |
| 05        | Oct. 31        | Self-supervised learning / 自己教師あり学習   | Learn how to build your speech system.                        |
| 06        | Nov. 7         | Foundation model / 基盤モデル              |                                                               |
| 07        | Nov. 14        | Sound recognition/synthesis / 環境音認識合成 |                                                               |
| 08        | Nov. 28        | Advanced topics / 発展的課題               |                                                               |
| 09        | Dec. 05        | Advanced topics / 発展的課題               | Learn how to build your speech system.                        |
|           |                |                                       |                                                               |
| 10        | Dec. 12        | Speech system building / 音システム構築      |                                                               |
| 11        | Dec. 19        | Speech system building / 音システム構築      |                                                               |
| 12        | Dec. 26        | Speech system building / 音システム構築      | Learn how to build your speech system.                        |
| 13        | Jan. 09        | Speech system building / 音システム構築      |                                                               |
| 14        | Jan. 23        | <b>Final presentation / 最終発表</b>      |                                                               |

# About this class

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- Language
  - Slides in English (w/ partial translation to Japanese, e.g., named entity)
  - Talk in Japanese (w/ simultaneous translation to English.)
    - Once translation finished for current slide, I will move onto next slide.
  - Either En or Ja is fine for your question.
- Grades
  - 50 / 100 ... final presentation
  - 50 / 100 ... final report
- Room
  - 14-202

# Final presentation & reports / 最終発表&レポート

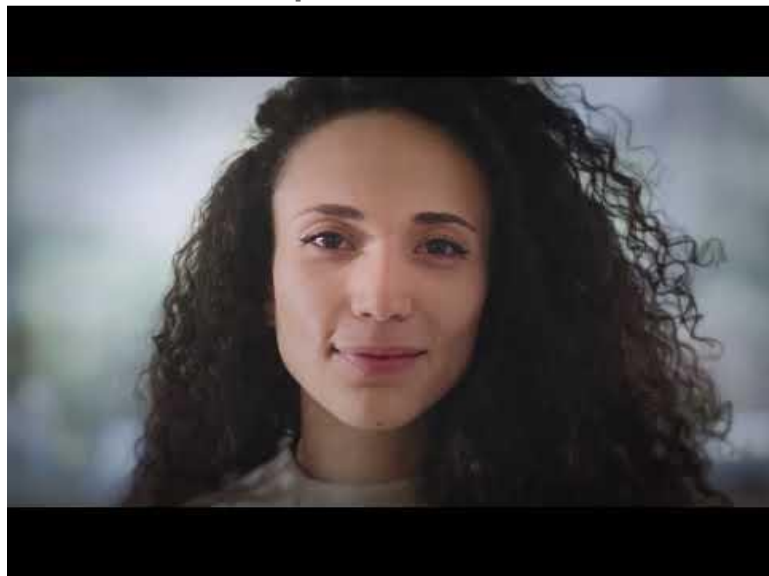
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- Perform a unique project that either
  - improves on state-of-the-art in speech-related {signal processing, machine learning} tasks,
  - applies {machine learning, signal processing} to a unique task,
  - investigates human behaviors in human–speech-AI interaction.
  - prototypes the future of human and speech AI.
- Presentation & report
  - Slides in English.
  - Talk in Japanese or English. (choose one you prefer.)
    - I will drive {Ja, En}-to-{En, Ja} simultaneous translation during presentation.
  - Reports in Japanese or English. (choose one you prefer.)

# Previous class: speech synthesis

## What happens in computers when artificially synthesizing speech

Text-to-speech / 音声合成

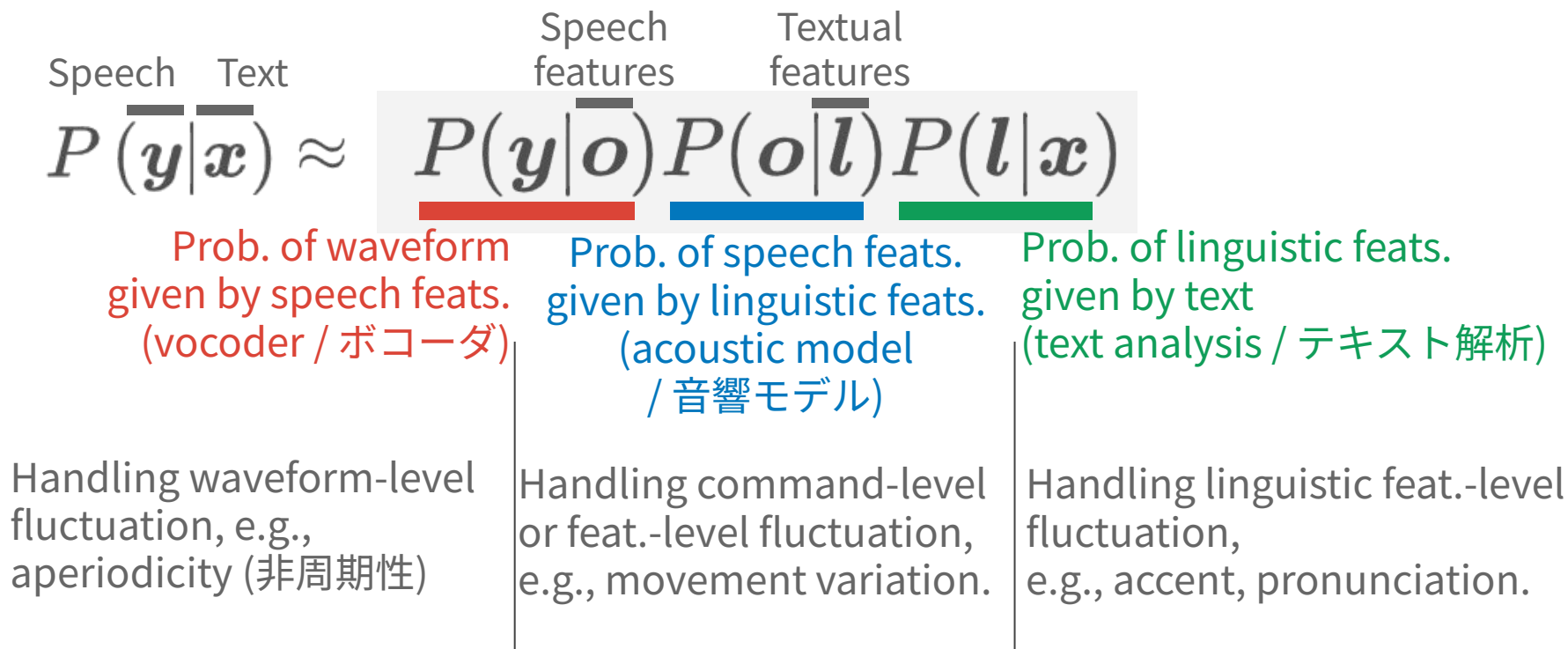


Singing voice synthesis / 歌声合成

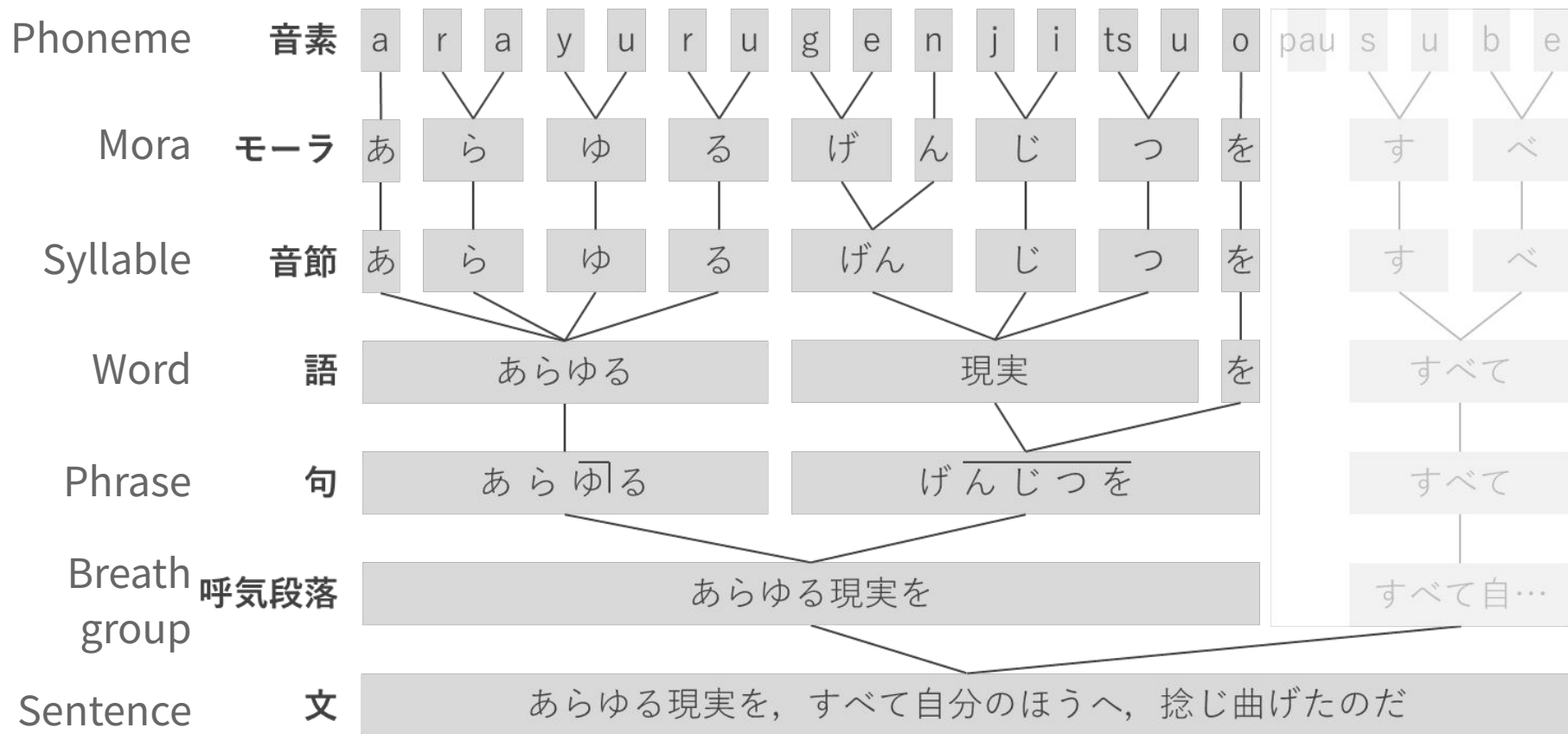


# Previous class: formulation of speech synthesis

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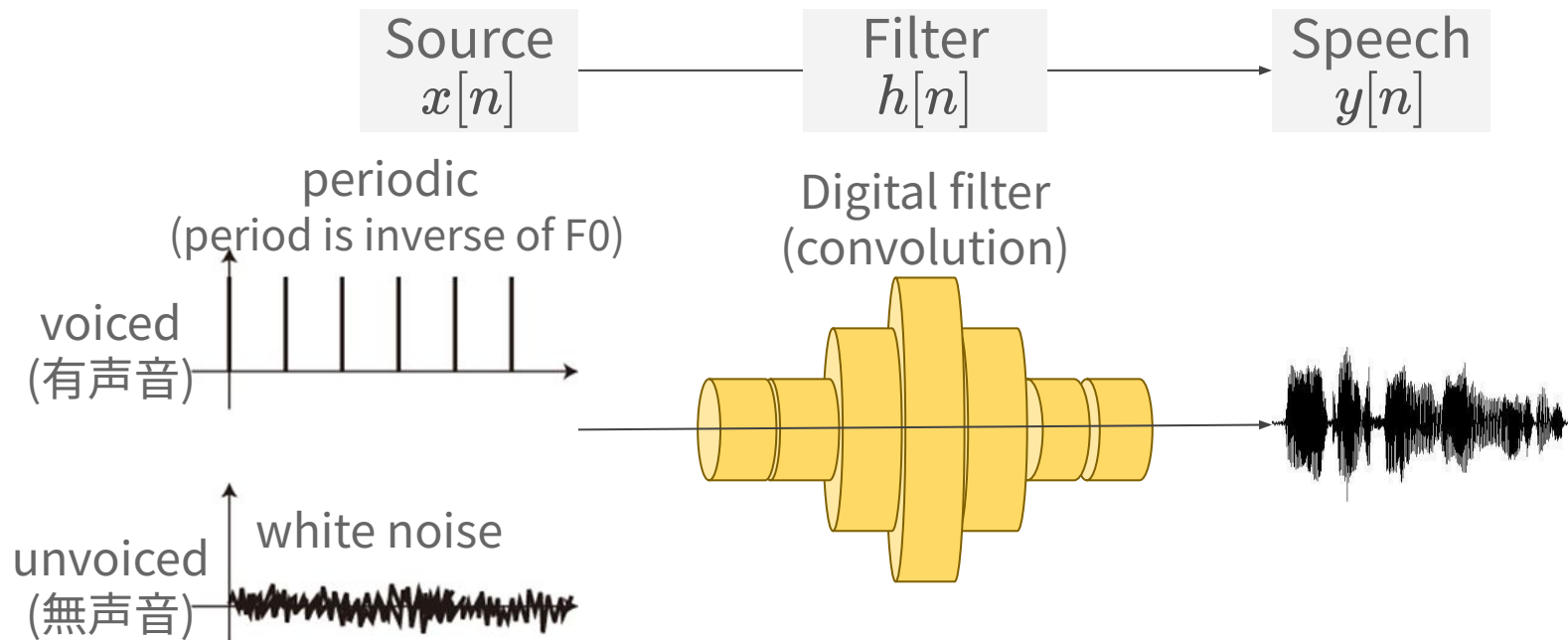


# Previous class: text analysis for Japanese



# Previous class: vocoder

Based on the source-filter theory, source and filter components are parameterized.



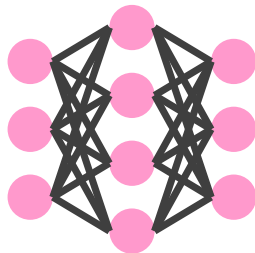
# Today's goal: acoustic models / 音響モデル

Modeling relationship between linguistic-acoustic features.

$$P(\overline{\mathbf{y}}|\overline{\mathbf{x}}) \approx P(\overline{\mathbf{y}}|\overline{\mathbf{o}}) \underbrace{P(\overline{\mathbf{o}}|\overline{\mathbf{l}})}_{\text{Prob. of speech feats. given by linguistic feats. (acoustic model / 音響モデル)}} P(\overline{\mathbf{l}}|\overline{\mathbf{x}})$$

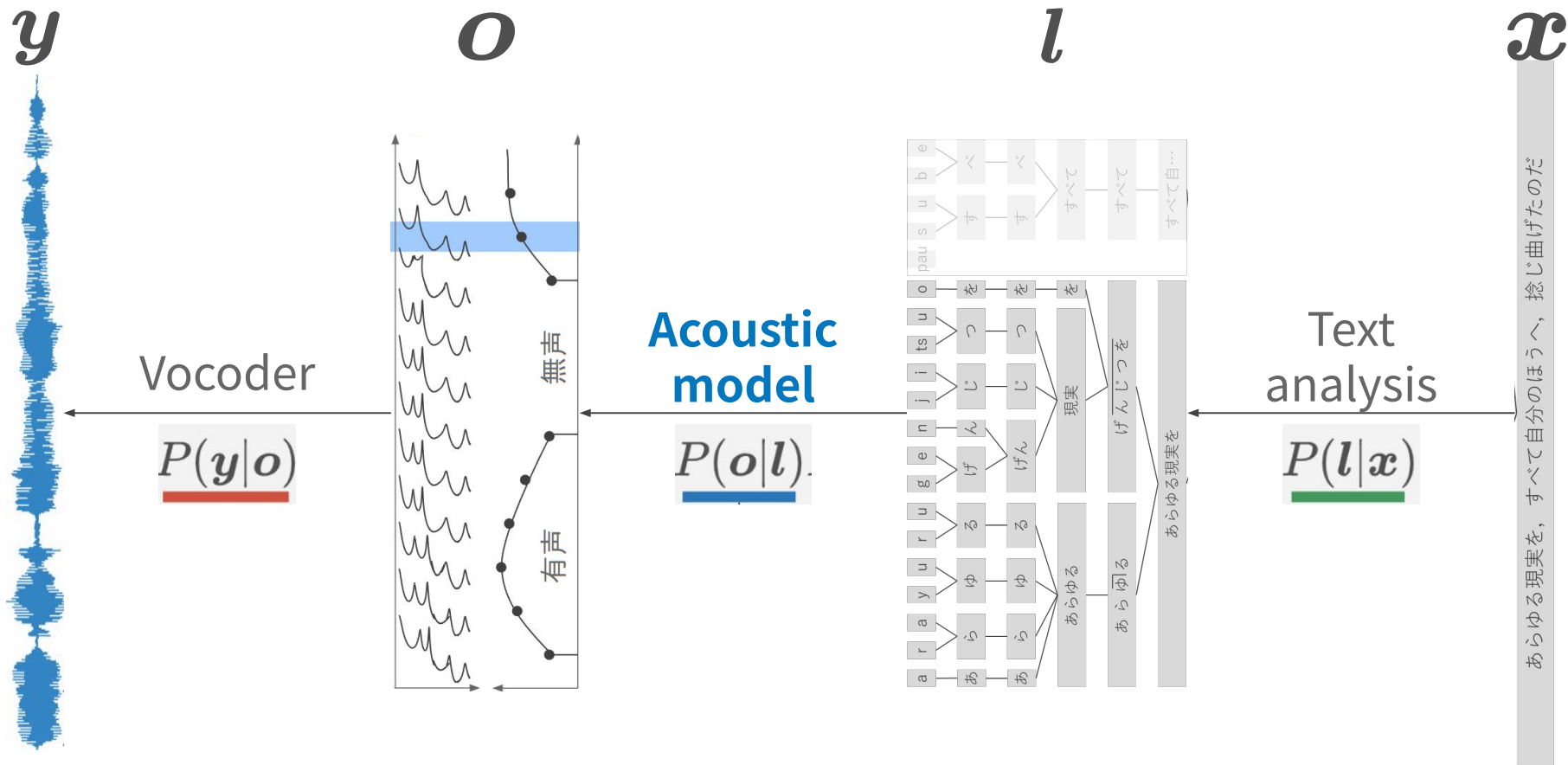
Speech features      Textual features

Speech      Text



# Acoustic model / 音響モデル

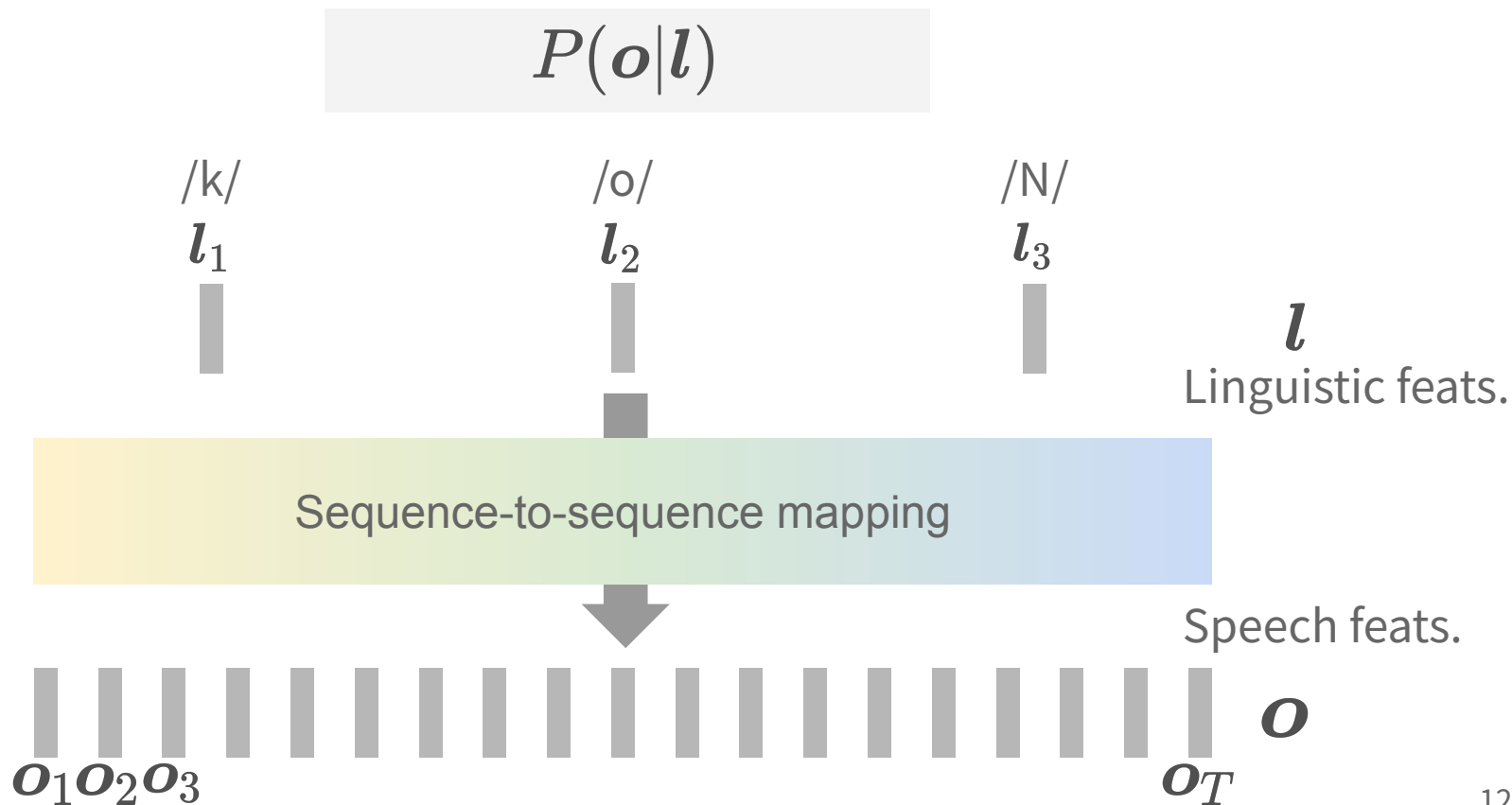
# Cascaded systems for speech synthesis / 音声合成におけるカスケード型システム



# Task definition of acoustic models

## / 音響モデルのタスク定義

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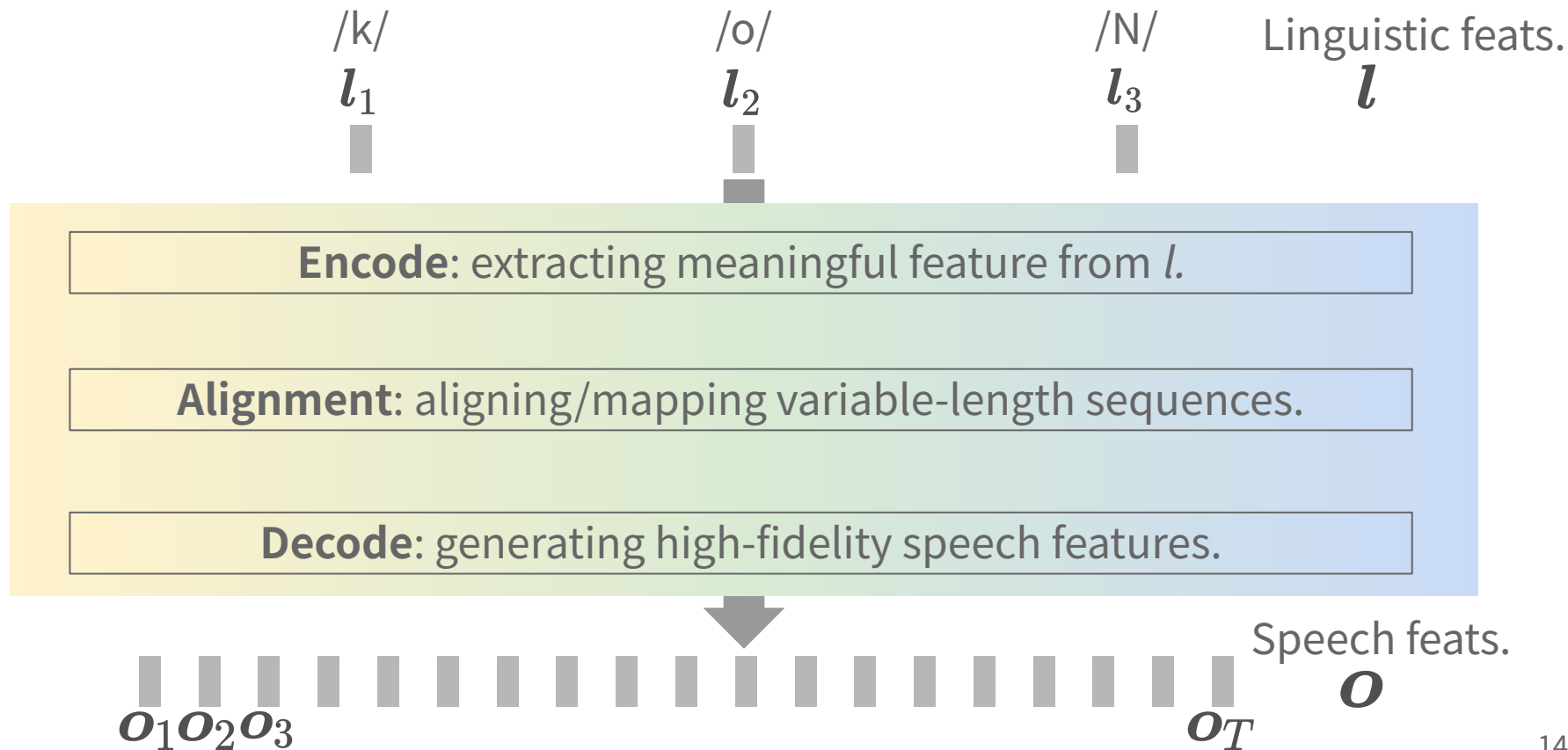
# Task definition of acoustic models

## / 音響モデルのタスク定義

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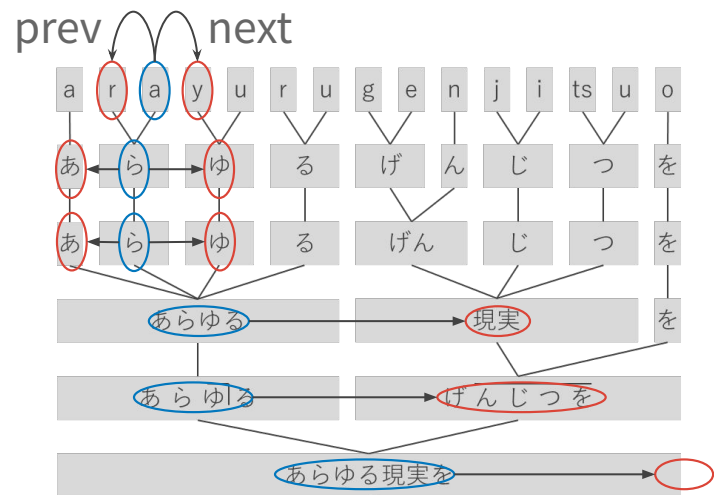
- length of ***l*** << length of ***O***
  - こんにちは (*konnichiwa*) -> **9** phonemes /k o N n i c h i w a/
  -  -> **200** frames (5 msec frame shift, 1 sec duration)
- Feature type:
  - Linguistic feats: (basically) discrete / 基本的には離散
  - Speech feats: (basically) continuous / 基本的には連続

# Into details: three inner modules



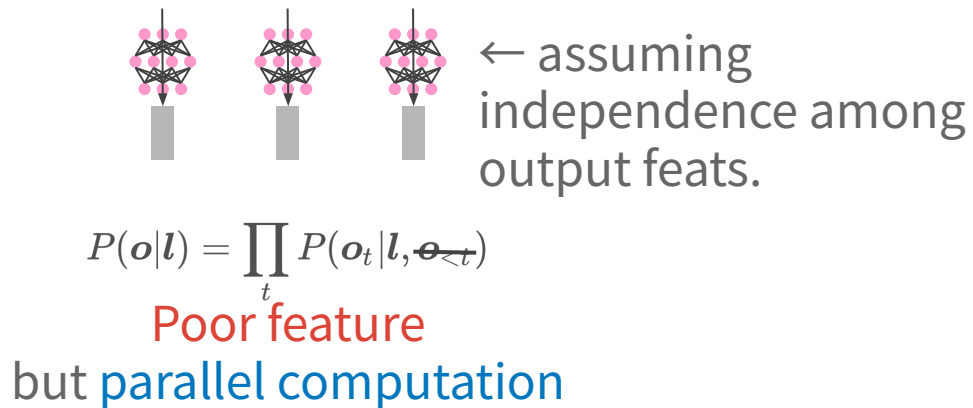
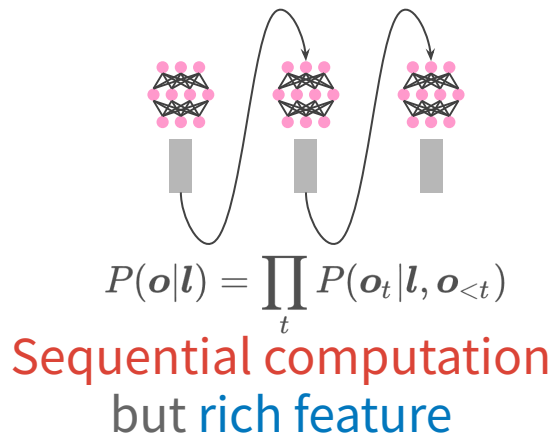
# Encode: extracting feature from linguistic features.

- Sequence-wise feature extraction
  - {previous, current, next} phonemes (e.g., co-articulation / 調音結合)
  - {previous, current, next} {mora, syllable, accent, word}
  - {positive, negative, question} sentence
  - position in a sentence
- Phonemes only is not enough as features.
  - Accent? Intonation? Other context?
  - Only phonemes are used to simplify my slide, but please do not forget that this setting is not enough.



# Decode: generating high-fidelity speech features

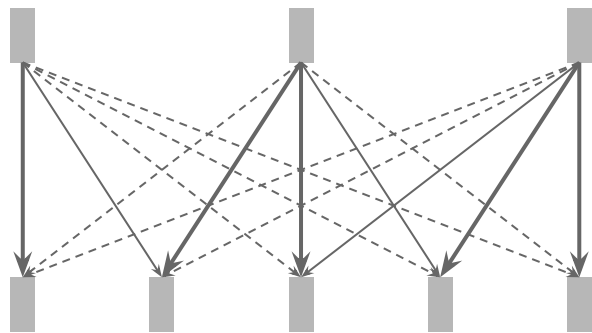
- Autoregressive vs. non-autoregressive / 自己回帰 vs. 非自己回帰



- Low-resolution vs. high-resolution / 低解像度 vs. 高解像度
  - High-resolution (e.g., spectrogram): difficult to predict but rich feature.
  - Low-resolution (e.g., vocoder feats): poor feature but easy to predict.

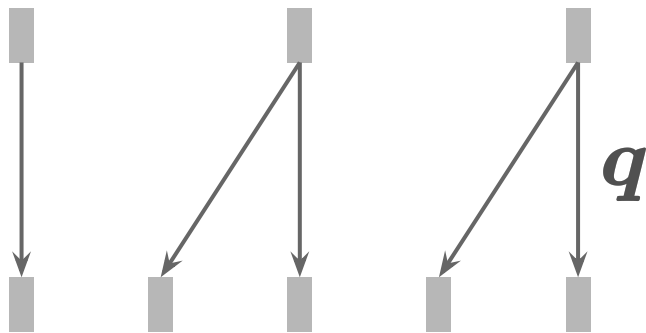
# Alignment: aligning/mapping variable-length seq.

- Soft alignment vs. hard alignment / ソフト vs. ハードアライメント



$$P(\mathbf{o}|\mathbf{l}) = \prod_t P(\mathbf{o}_t|\mathbf{l})$$

Consider  
all inputs.



$$P(\mathbf{o}|\mathbf{l}) = \prod_t \underbrace{P(\mathbf{o}_t|\mathbf{l}, \mathbf{q})}_{\text{Hardly aligns feats.}} \underbrace{P(\mathbf{q}|\mathbf{l})}_{\text{Approximate the alignment with optimum}}$$

Hard computation but rich feature

Poor feature but light computation

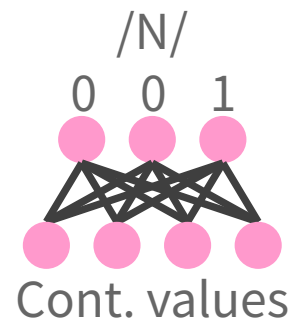
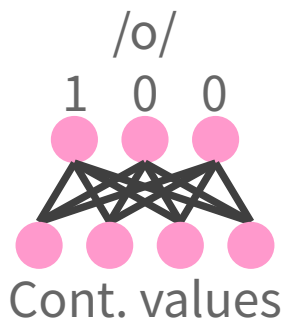
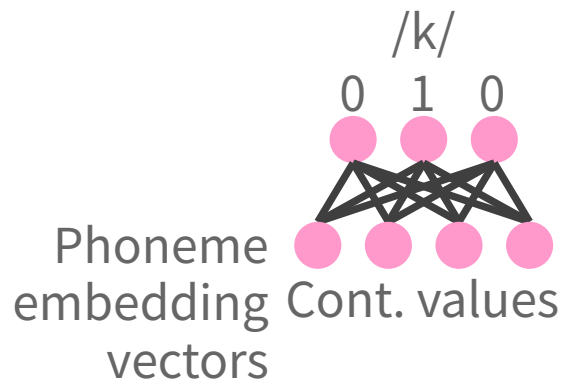
# Key module: word embedding layer / 単語埋め込み層



(Fundamentals of deep learning is out of scope of this class. Please study in advance.)<sup>18</sup>

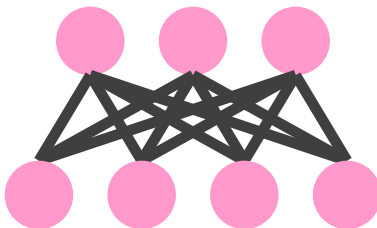
# Phoneme embedding / 音素埋め込み層

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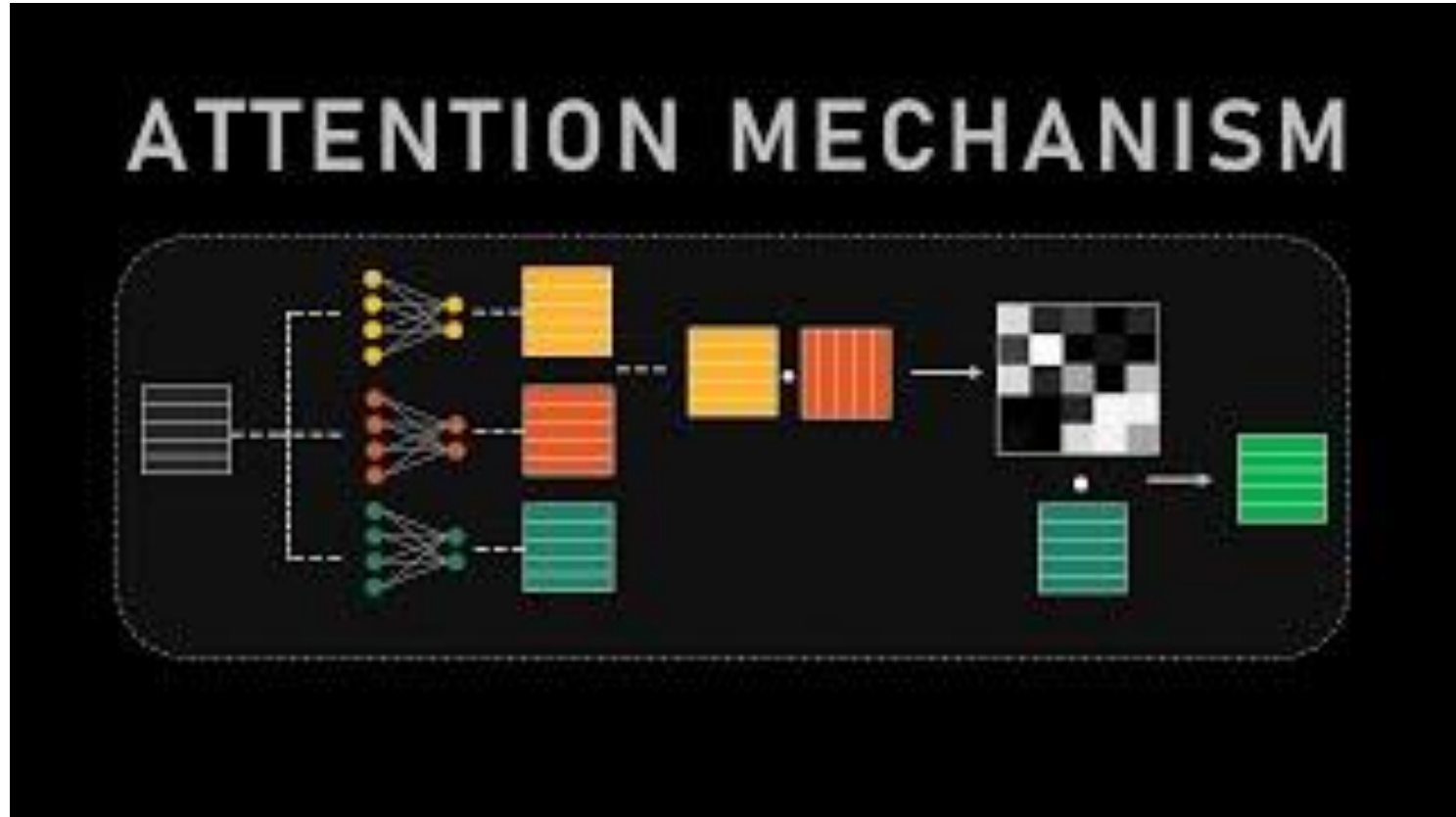
Size: # of phoneme candidates.

Size: user-definable (e.g., 512)



# Key module: self-attention and Transformer

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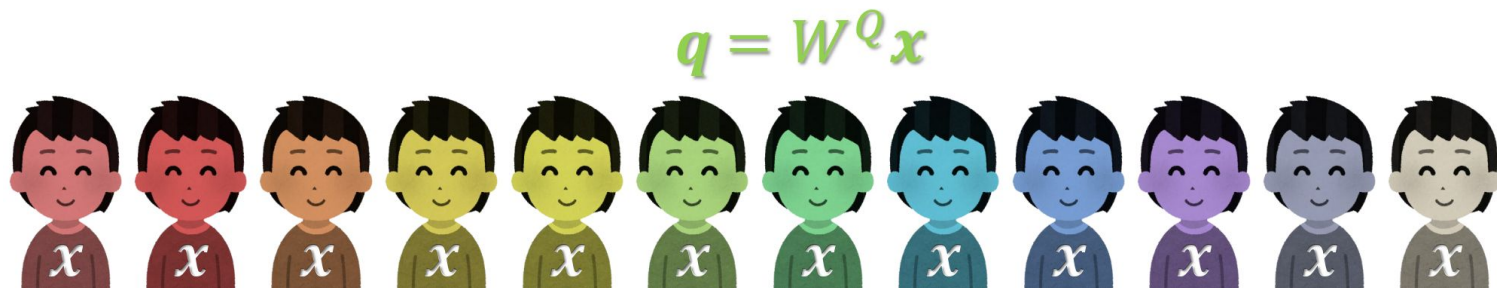
(Fundamentals of deep learning is out of scope of this class. Please study in advance.)<sup>20</sup>

# Self attention-based feature extraction

このベクトル  に注目して考える

1. 他のベクトルを「検索」するためのクエリ  $q = W^Q x$  を計算

Compute **query vector(s)** from input to search other vectors.



# Self attention-based feature extraction

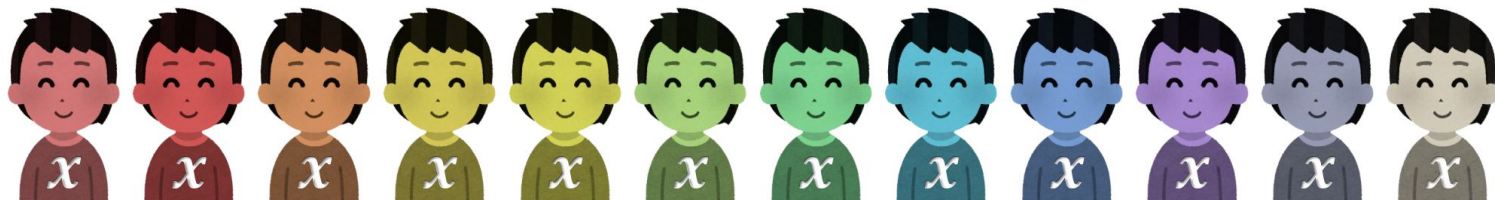
このベクトル  に注目して考える

2. 「類似検索」対象としての各ベクトルのキー  $k = W^K x$  を計算

Compute **key vector(s)** from input to be searched from other vectors.

$k$   $k$   $k$   $k$   $k$   $k$   $k$   $k$   $k$   $k$   $k$

$q$



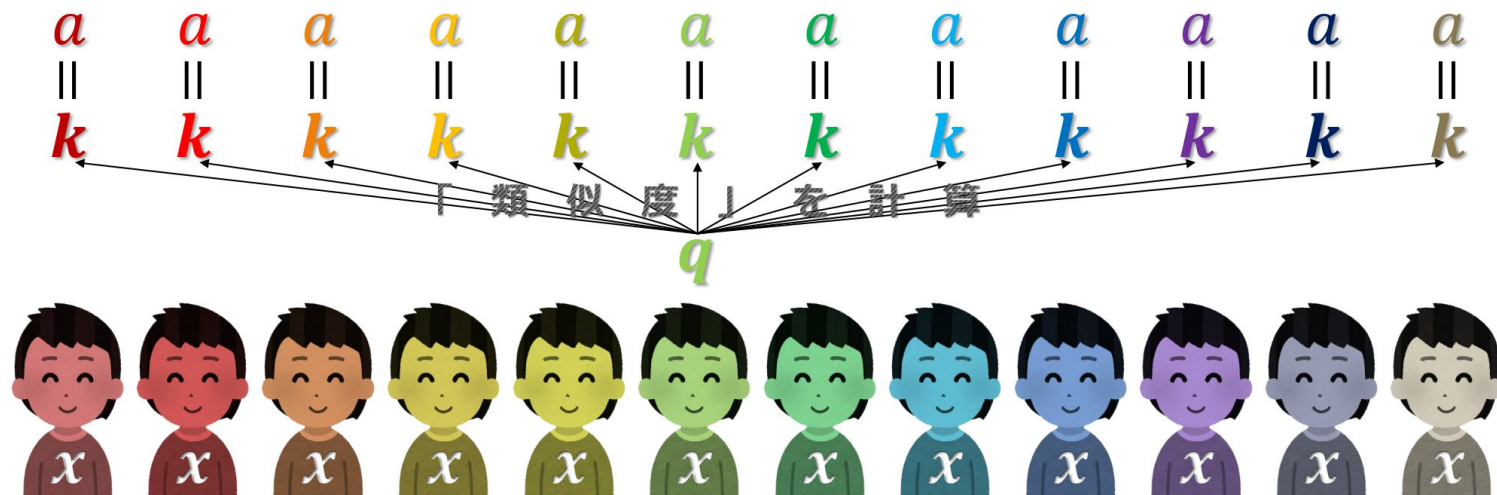
# Self attention-based feature extraction

このベクトル  に注目して考える

3. クエリとキーの内積を次元数 $d$ とsoftmaxで正規化した「類似度」

$a = \text{softmax}(q^T k / \sqrt{d})$  を計算

Compute **vector similarity** (= **attention**) between query and key.

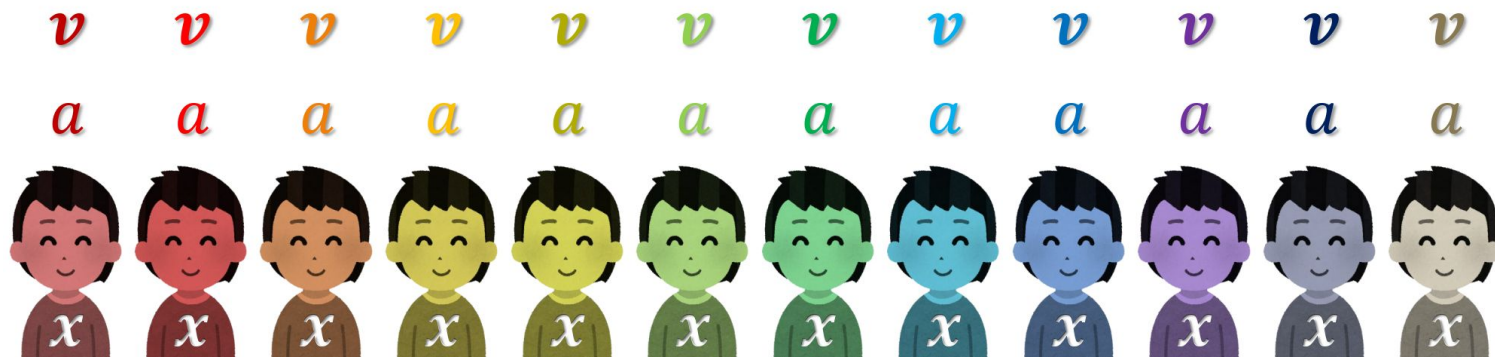


# Self attention-based feature extraction

このベクトル  に注目して考える

4. 各ベクトルの代表値  $v = W^v x$  を計算

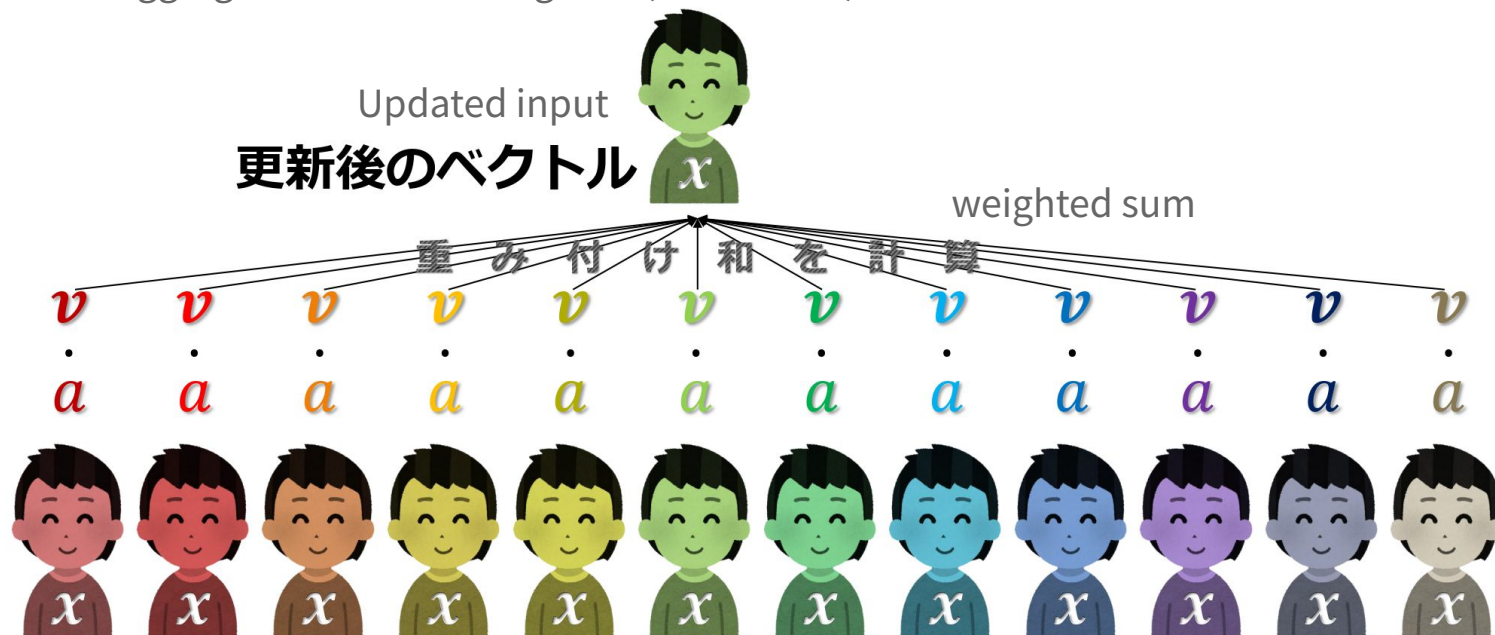
Compute **value vector(s)** from input to represent meaningful features.



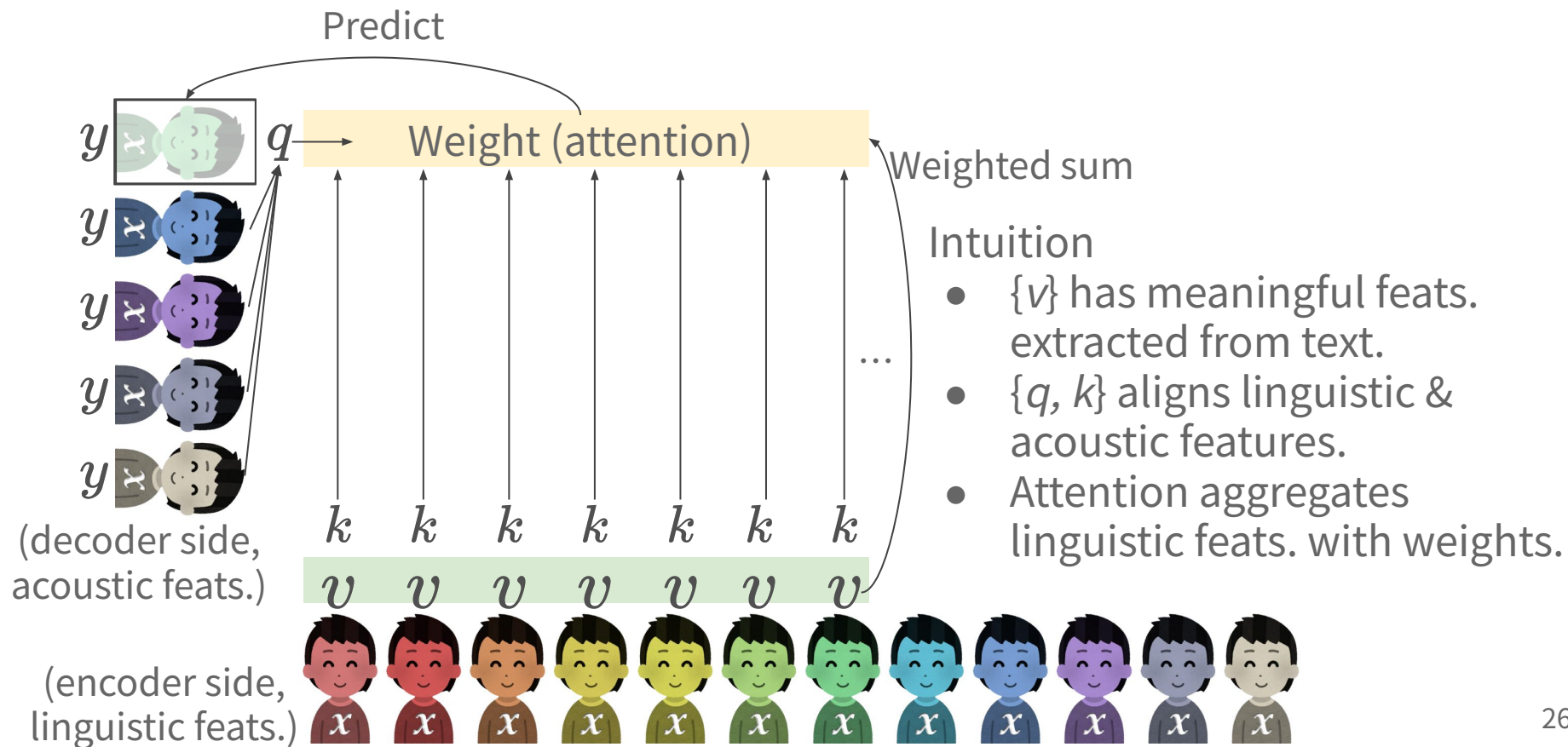
# Self attention-based feature extraction

このベクトル  に注目して考える

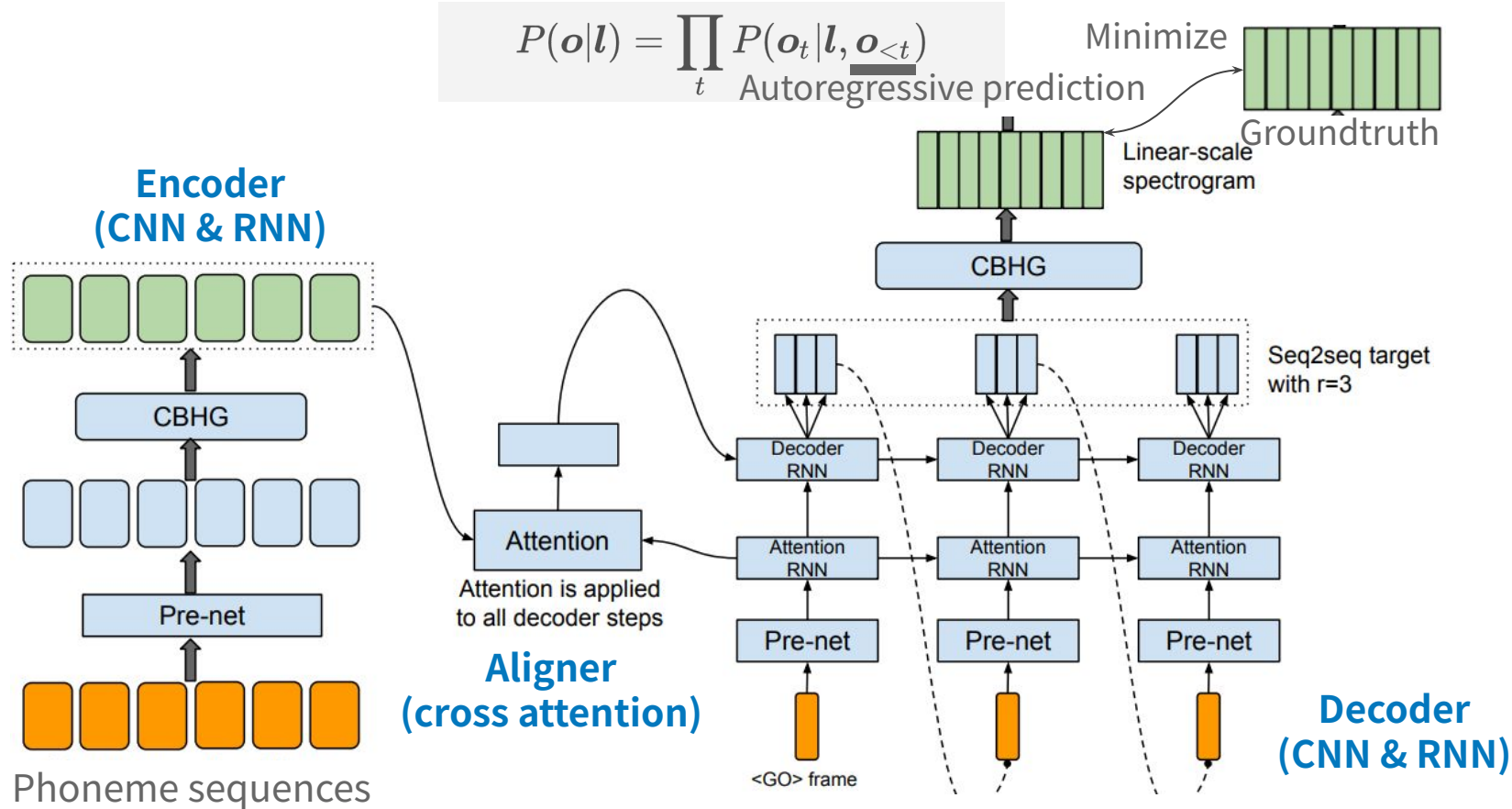
5. 類似度 $a$ で重みづけした和 $\sum av$ を計算→ベクトル  に加算 (residual接続有)  
Aggregate all of the “weighted (= attention)” value vectors.



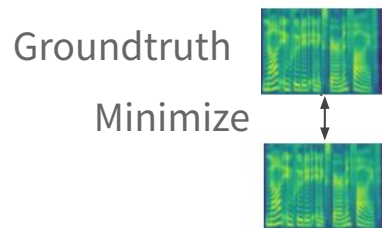
# Cross-attention



# Tacotron: pioneer of end-to-end TTS

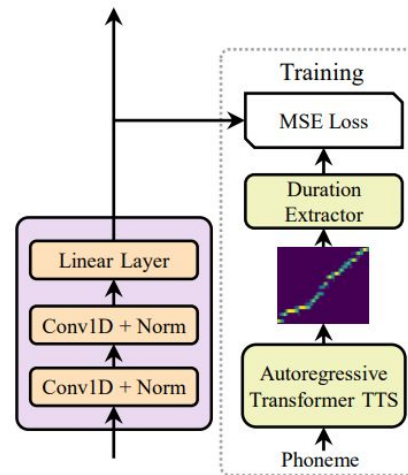
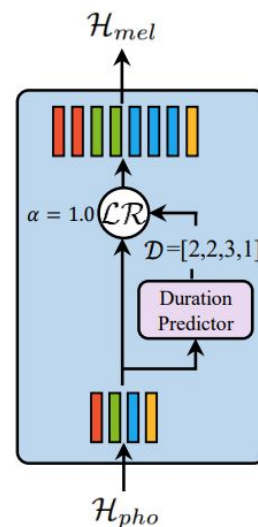
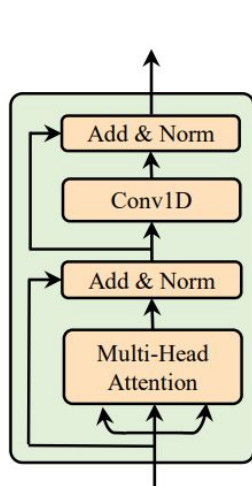
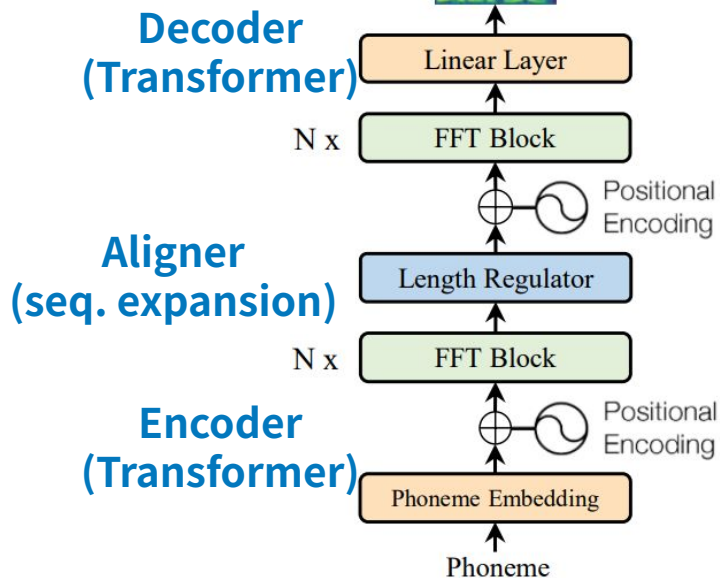


# FastSpeech: duration-informed TTS



$$P(\mathbf{o}|\mathbf{l}) = \prod_t \underbrace{P(\mathbf{o}_t|\mathbf{q}, \mathbf{l})}_{\text{Feature prediction}} \underbrace{P(\mathbf{q}|\mathbf{l})}_{\text{Alignment}}$$

Feature prediction  
under given alignment



(a) Feed-Forward Transformer

(b) FFT Block

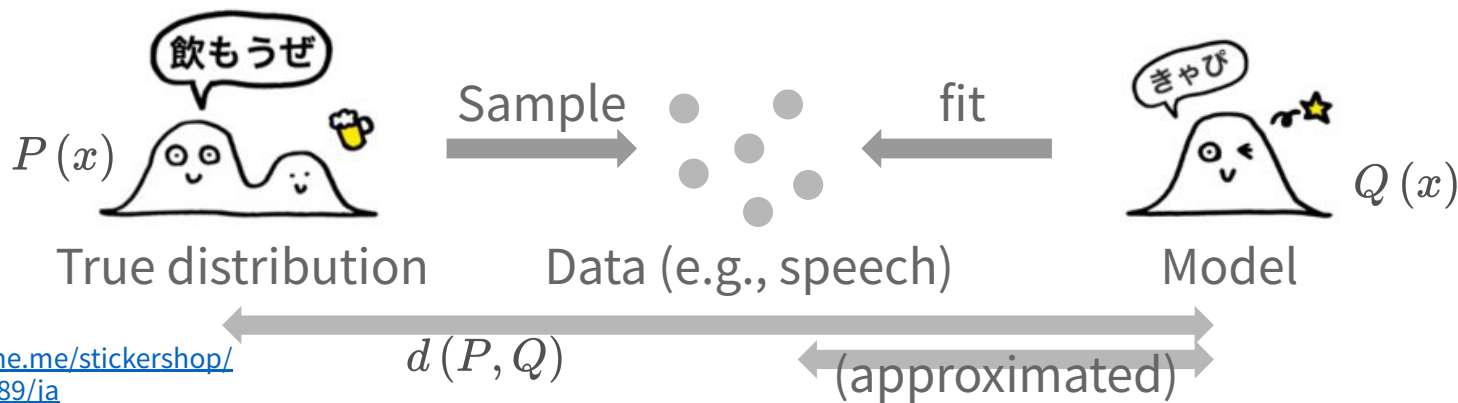
(c) Length Regulator

(d) Duration Predictor

# Deep generative models in acoustic models / 音響モデルにおける深層生成モデル

# Maximum likelihood and Kullback-Leibler divergence / 最尤推定とカルバックライブラー距離

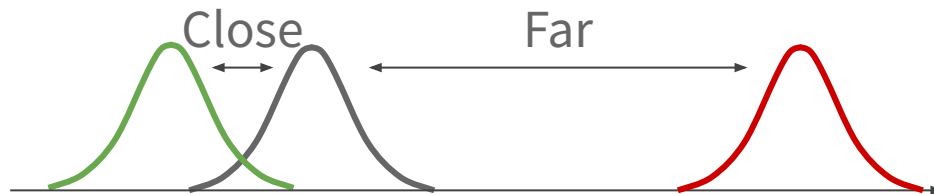
- Data follows the true probability distribution, which is not observable for us; we hypothesize that the data follows a known distribution. / データはある真の確率分布に従うが、我々はそれを知ることはできない。故に、データが何らかのモデルに従うと仮定する。
  - $P(x)$ : true distribution (unobservable) / 真の確率分布 (未知)
  - $Q(x)$ : distribution we hypothesized / 我々が仮定した確率分布
  - $d(P, Q)$ : distance betw. two distr. / 2つの確率分布の間の距離



# Divergence between probability distributions / 確率分布の間の距離

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- Statistical distance between probability distributions.



- Many variants.
  - f-divergence
    - Kullback-Leibler divergence (KLD), Jensen-Shannon divergence (JSD), ...
  - Bregman divergence
  - etc.

# Maximum likelihood and Kullback-Leibler divergence / 最尤推定とかるバックライブラー距離

- Minimizing Kullback Leibler (KL) divergence

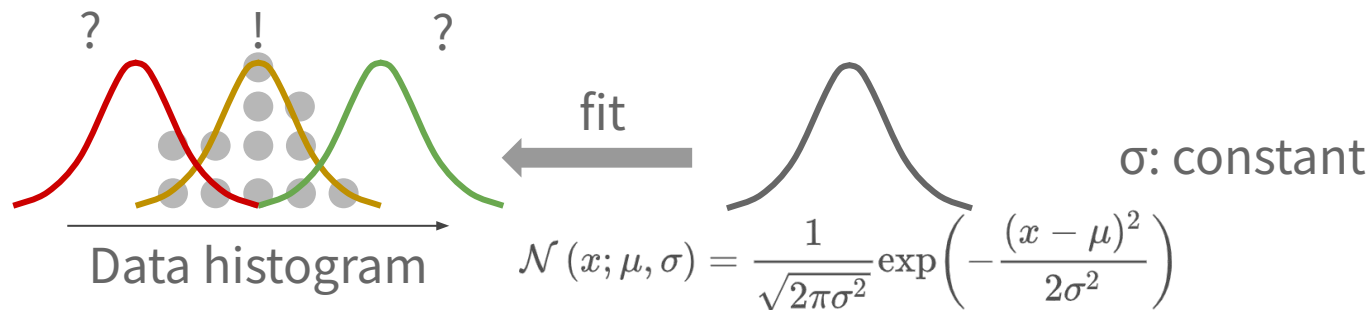
$$d(P, Q) = \text{KL}(P, Q) = \int P(x) \log \frac{P(x)}{Q(x)} dx = \underbrace{\int P(x) \log P(x) dx}_{\text{Constant / 定数}} - \underbrace{\int P(x) \log Q(x) dx}_{\substack{E[\log Q(x)] \\ \text{Expectation / 期待値}}}$$

- The second term is equivalent maximum likelihood estimation.
  - I.e., estimating most likely parameters for all the data.

$$- \int P(x) \log Q(x) dx$$

# Example

- Assuming the data follows the Gaussian distribution with fixed variance. / データが「定数標準偏差の正規分布」に従うと仮定
  - Mean parameter needs to be estimated. / 求めたいのは平均のみ.



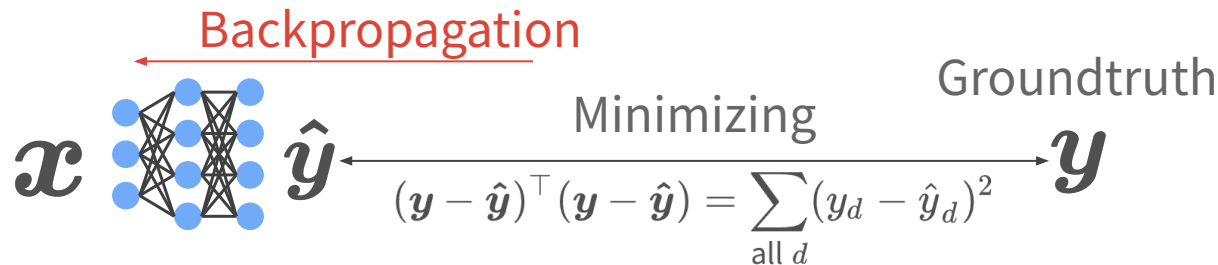
- Logarithm of likelihood / 尤度の対数

$$-\log \prod_{\text{all } x} \mathcal{N}(x; \mu, \sigma) \propto \sum_{\text{all } x} (x - \mu)^2 + \text{Const}$$

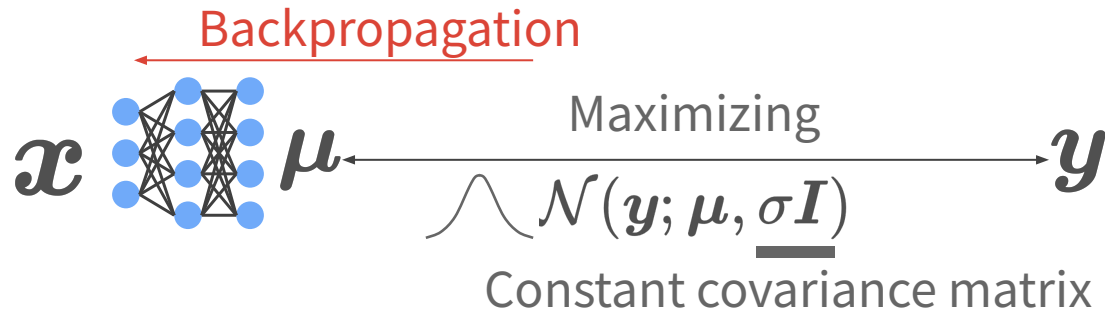
Minimizing mean squared error = Estimating mean of Gaussian.  
/ 二乗誤差の最小化 = 正規分布の平均の推定

# Mean squared error (MSE) in deep learning

- You often use the loss in deep learning.



- When using MSE, you are estimating mean (vector) of the Gaussian distribution with the constant variation (& covariance).



# Loss functions correspond to probability distributions.

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- Gaussian distribution with constant variance

$$\mathcal{N}(x; \mu, \sigma_{\text{const.}}) \longrightarrow (x - \mu)^2$$

- Gaussian distribution

$$\mathcal{N}(x; \mu, \sigma) \longrightarrow (x - \mu)^2 - 2\sigma^2 + \log \sigma$$

- Log-normal distribution / 対数正規分布

$$\mathcal{N}(\log x; \mu, \sigma) \longrightarrow (\log x - \mu)^2 - 2\sigma^2 + \log \sigma$$

- Laplace distribution with constant scale parameter  $b$

$$\frac{1}{2b_{\text{const}}} \exp\left(-\frac{|x - \mu|}{b_{\text{const}}}\right) \longrightarrow |x - \mu| \quad (\text{L1 norm})$$

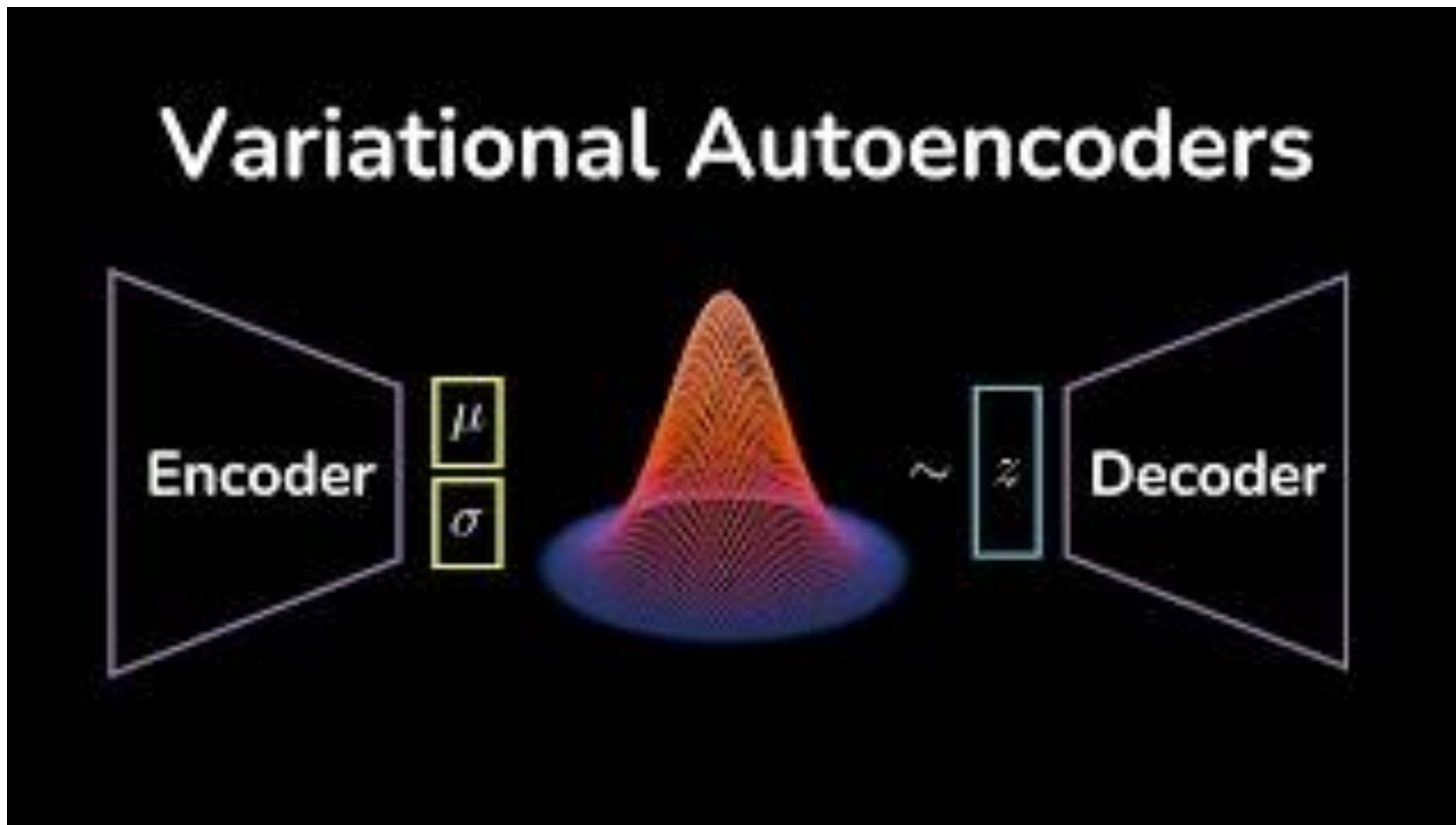
- von Mises distribution with constant parameter  $\beta$

$$\frac{1}{2\pi I_0(\beta_{\text{const}})} \exp[\beta_{\text{const}} \cos(x - \mu)] \longrightarrow -\cos(x - \mu) \quad (\text{negative cosine similarity})$$

(Deep) generative models assume probability distribution.

# Variational autoencoder / 変分オートエンコーダ

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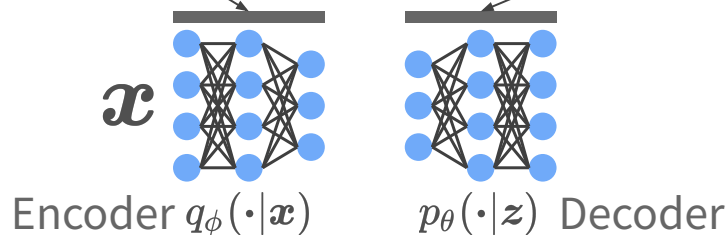
Can be seen as transforming shapes of probability distributions.

# Variational autoencoder / 変分オートエンコーダ

Training objective

$$\text{KL}(q_\phi(\mathbf{z}|\mathbf{x}), p_\theta(\mathbf{z})) - E[\log p_\theta(\mathbf{x}|\mathbf{z})]$$

(simplified notation)



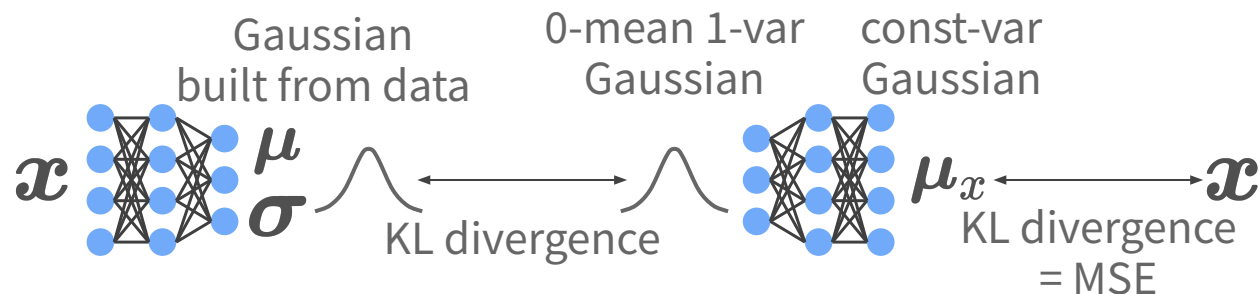
- Assumption

- $q_\phi(\cdot|\mathbf{x})$  is the Gaussian distr.
- $p_\theta(\cdot|\mathbf{z})$  is the Gaussian distr. with constant variable.
- $p_\theta(\mathbf{z})$  is the 0-mean unit-variance Gaussian distribution.

# Variational autoencoder / 変分オートエンコーダ

Training objective

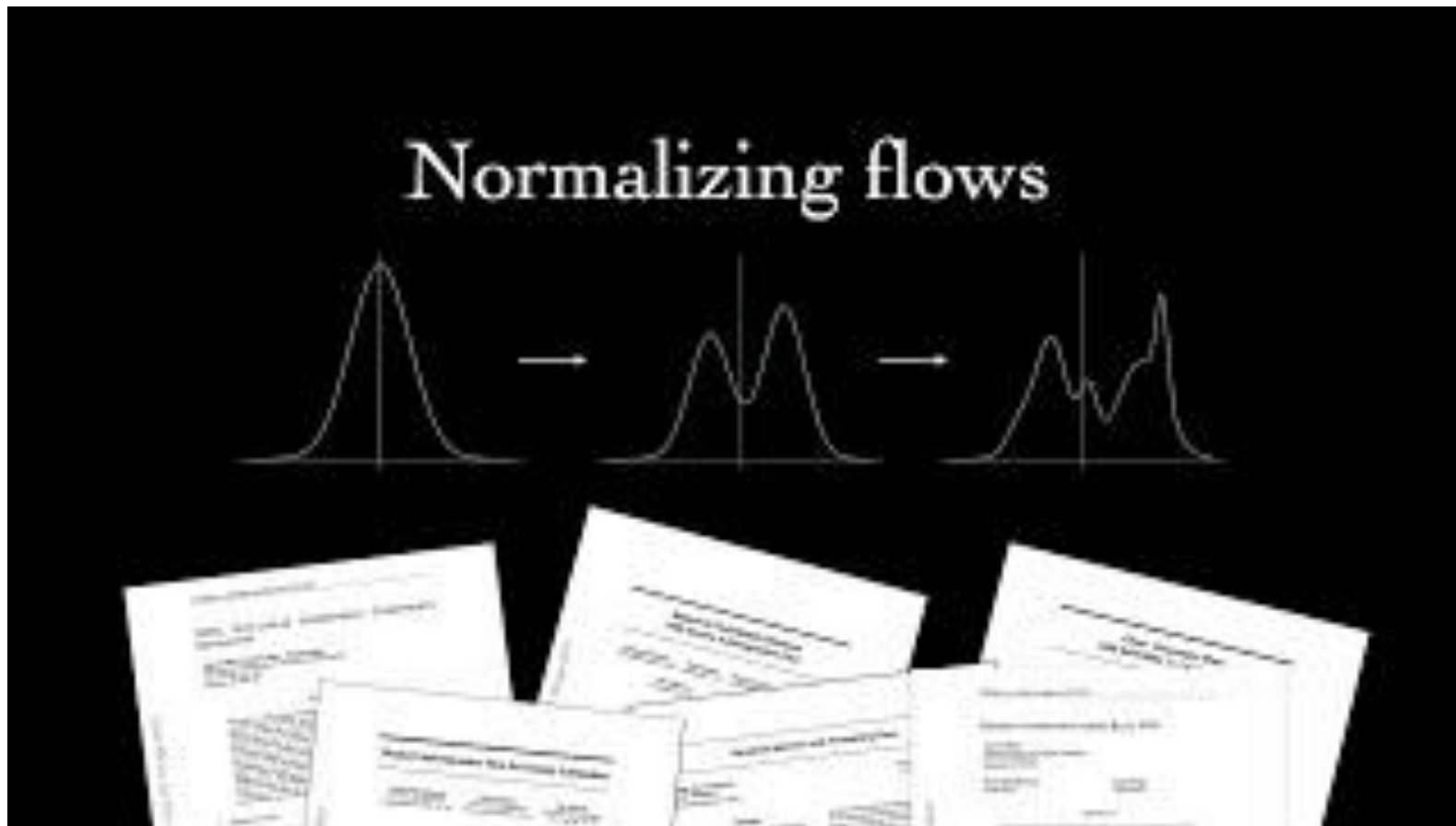
$$\underline{\text{KL}(q_\phi(\mathbf{z}|\mathbf{x}), p_\theta(\mathbf{z}))} - \underline{E[\log p_\theta(\mathbf{x}|\mathbf{z})]}$$



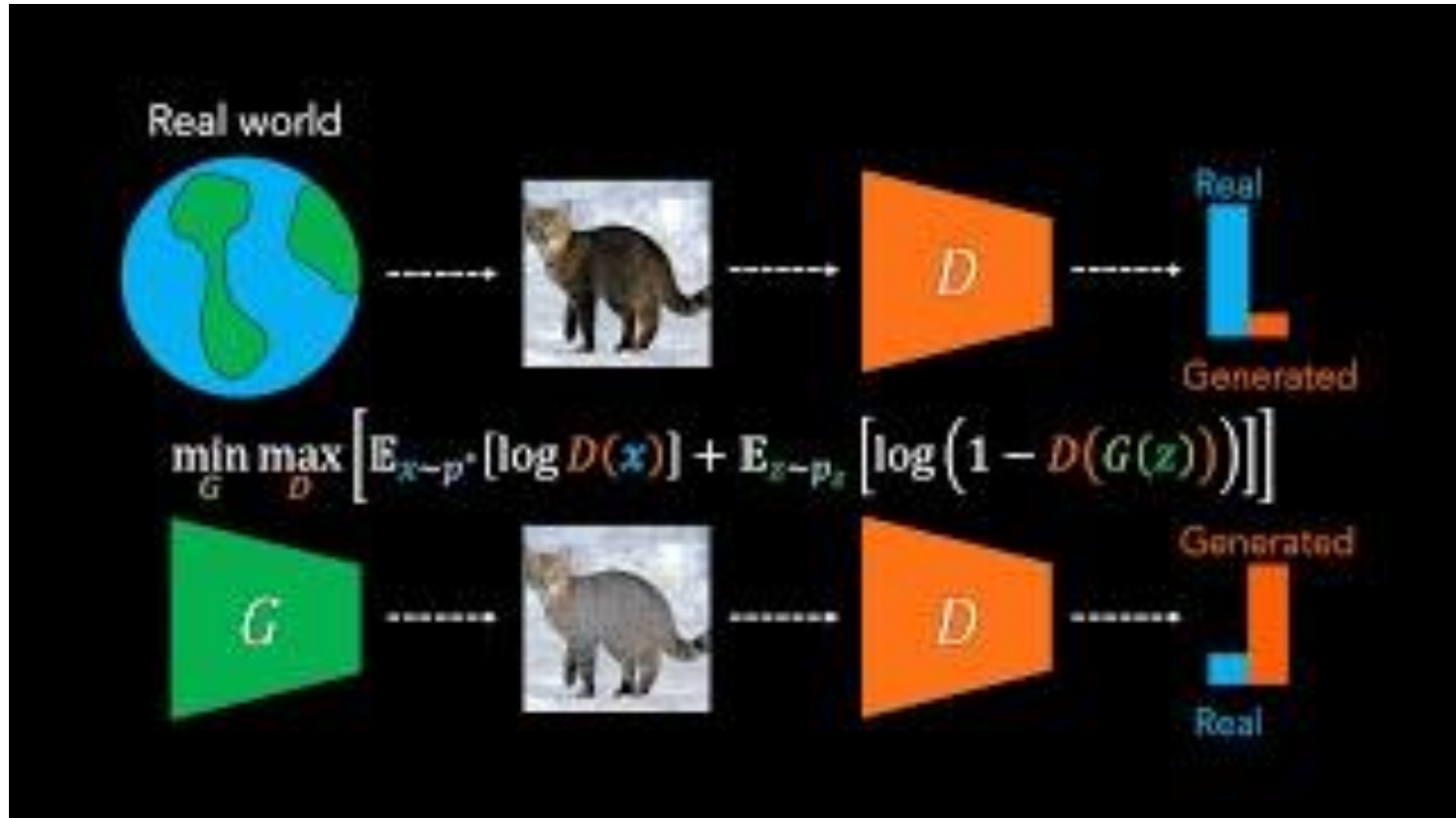
- Encoder transforms data distribution to 0-mean 1-var Gaussian.
- Decoder transforms 0-mean 1-var Gaussian to data distribution.
- Intuitively, non-linear version of principal component analysis.

Normalizing flow (abstract): invertible transformation between data distribution and prior distribution.

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# GAN (generative adversarial network) / 敵対的生成ネットワーク



Can be seen as transforming known distribution to data distribution.

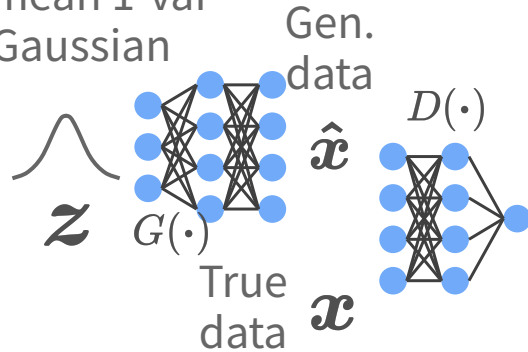
# GAN (generative adversarial network) / 敵対的生成ネットワーク

## Training objective

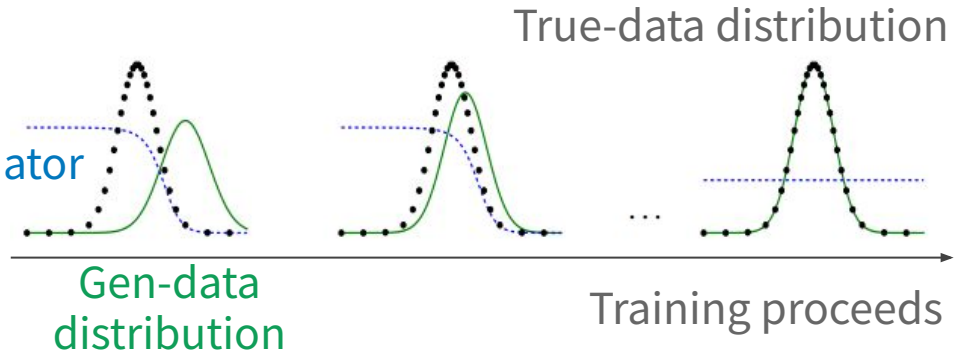
$$\min_G \max_D V(D, G) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}(\mathbf{z})} [\log(1 - D(\hat{\mathbf{x}}))].$$

let  $G(\cdot)$  fool   let  $D(\cdot)$  distinguish   “1” for true data   “0” for gen. data

0-mean 1-var  
Gaussian



Discriminator

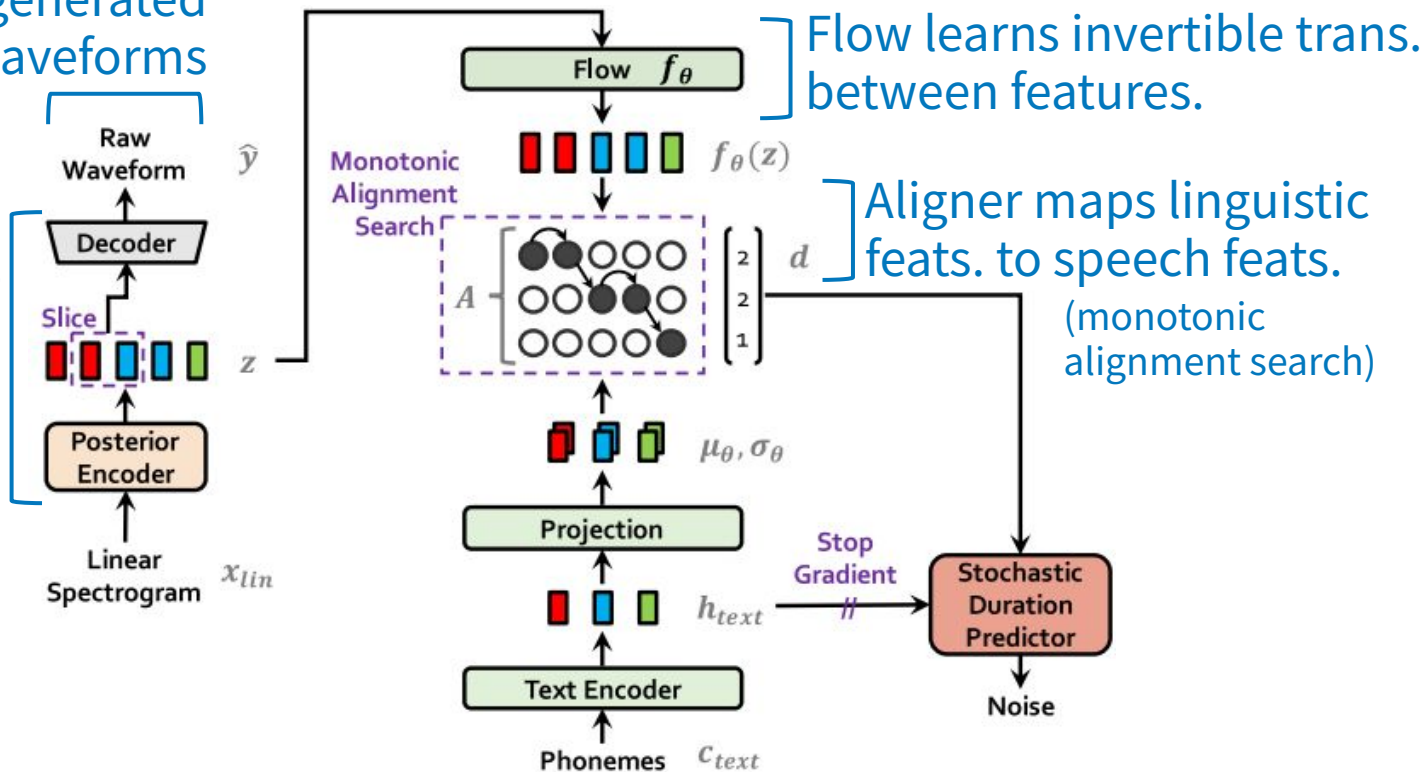


- GAN minimizes the Jensen-Shannon divergence (JSD).
  - JSD = 0 means two distributions are completely the same.
  - -> Generator can transform known distribution to data distribution.

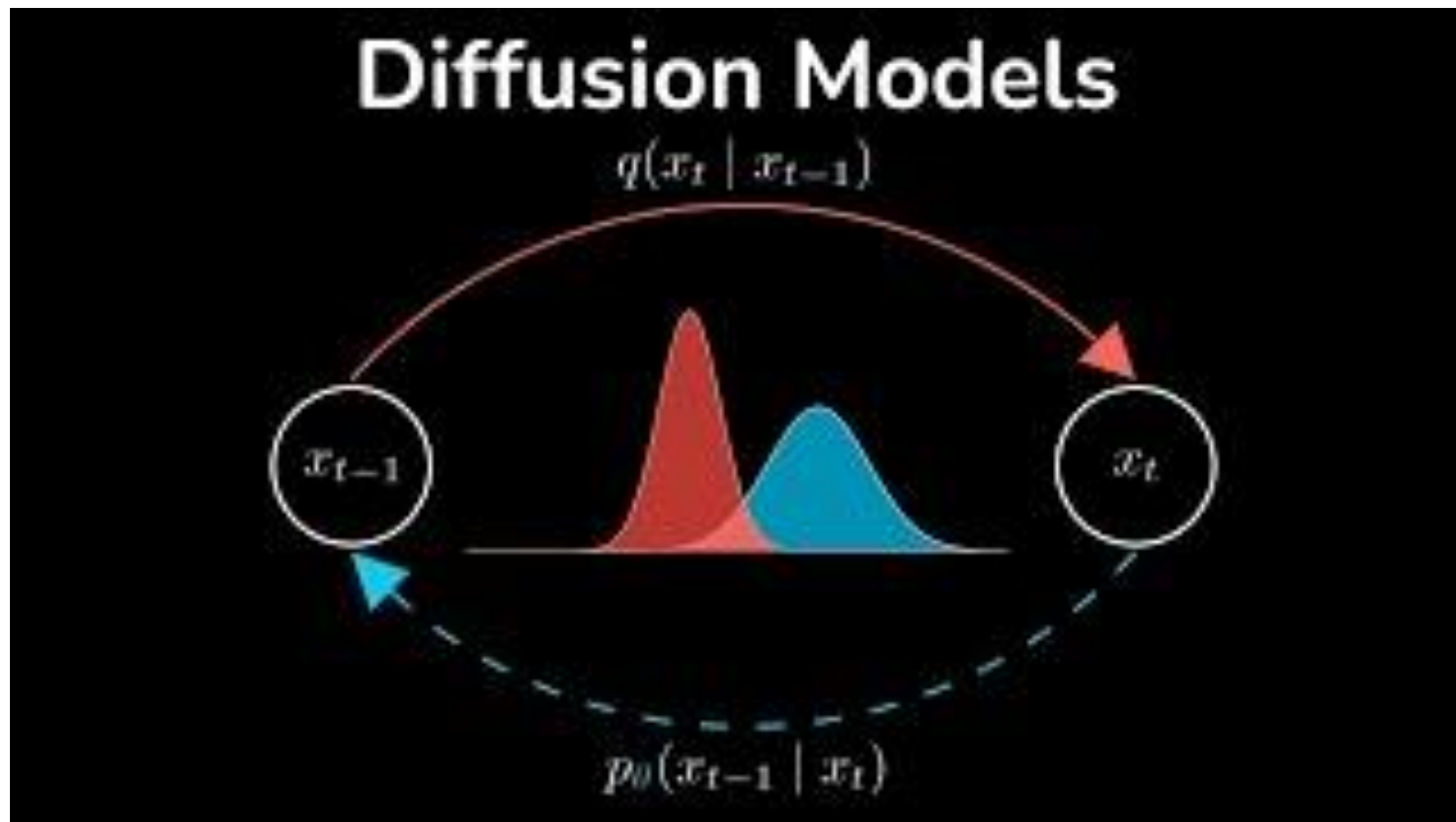
# VITS combining VAE, Flow, and GAN

GAN between generated and natural waveforms

VAE encodes spectrogram to known distribution.



# Diffusion model / 拡散モデル



Can be seen as transforming known distribution to data distribution.

# Diffusion model / 拡散モデル

## Training objective

$$\mathbb{E}_q \left[ \underbrace{D_{\text{KL}}(q(\mathbf{x}_T | \mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{L_T} + \sum_{t>1} \underbrace{D_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t))}_{L_{t-1}} \underbrace{- \log p_{\theta}(\mathbf{x}_0 | \mathbf{x}_1)}_{L_0} \right]$$

Artificial denoising (noise removing)

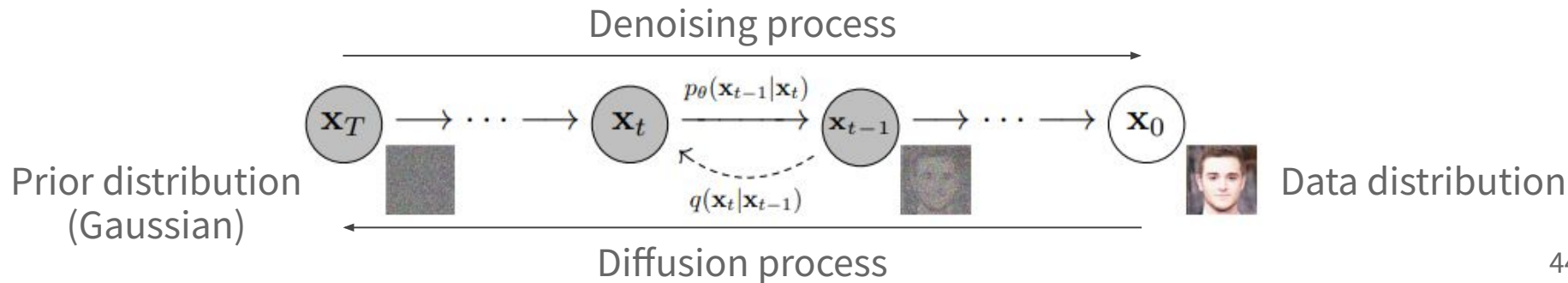
-> Gaussian with constant variance

Learnable denoising (noise removing)

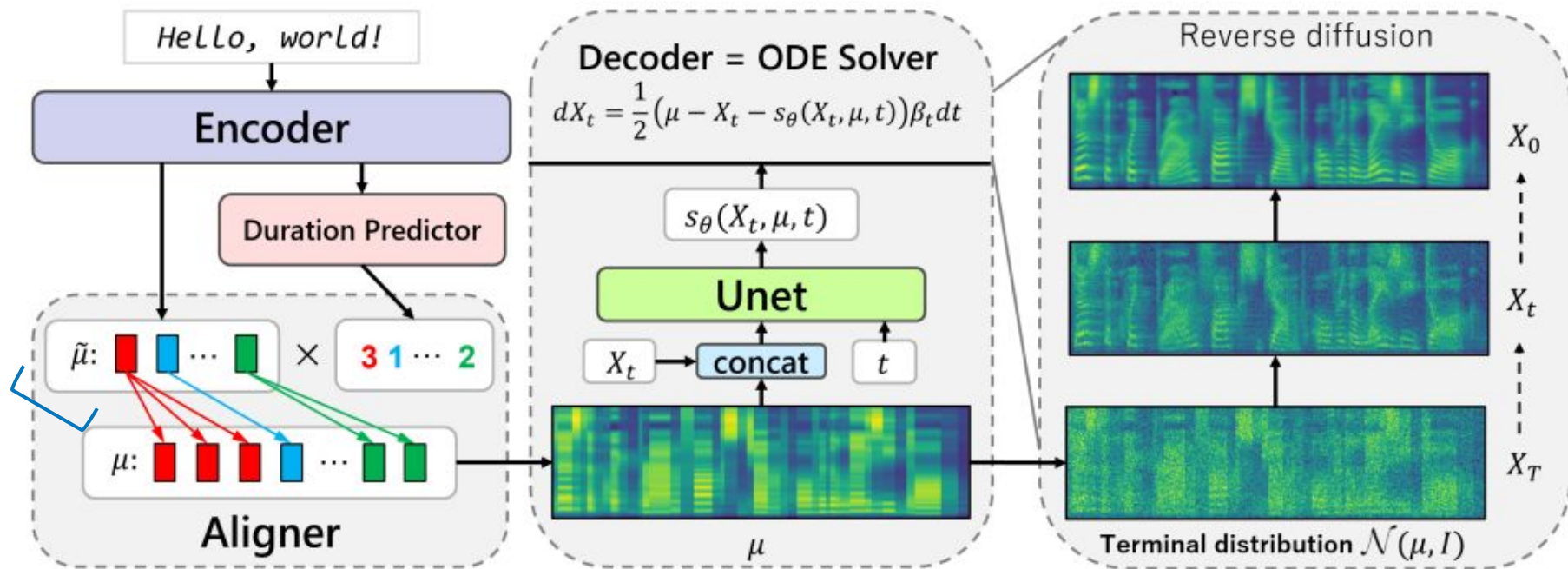
-> Gaussian with constant variance

- In summary

- Starting from prior distr., the generator iterates to remove noise of data.
- KL distance between two Gaussians with const. var. -> **MSE (-like loss)!**



# GradTTS: diffusion-based speech synthesis



Predicting mean vectors of phoneme-specific prior for diffusion models.  
(Intuitively, average spectrum of each phoneme.)

## Conclusion / まとめ

# Conclusion

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- Acoustic model / 音響モデル
  - Mapping linguistic feats. to speech feats.
  - Variable-length mapping
- Deep generative model / 深層生成モデル
  - Generative model learns probability distribution.
    - Minimizing divergence (distance) between true distribution and model's distribution.
  - Gaussian, VAE, Flow, GAN, Diffusion
  - Deep generative models in acoustic models, e.g., VITS, GradTTS

# MFA workshop in Keio (if you have interests, let's join!)

2025/11/06 13:00-18:00

## Montreal Forced Aligner (MFA) Workshop in Keio

Montreal Forced Aligner ワークショップ in 慶應義塾大学

Yagami Campus, Keio University / 慶應義塾大学 矢上キャンパス

<https://sites.google.com/keio.jp/mfa-workshop-keio>

Registration / 参加登録

### Special workshop for MFA! / MFA の特別なワークショップ

The Montreal Forced Aligner (MFA) is a powerful, open-source tool that align speech and text. It delivers high-accuracy, reproducible, and language-flexible forced alignment. We are pleased to welcome the main maintainer of the MFA for a special workshop.

Montreal Forced Aligner (MFA) は、オープンソースのテキスト音声アライメントツールです。本ツールは、その品質と再現性の高さ、そして言語依存性の低さから、世界中で使われています。このたび、MFA の主開発者の方をお迎えして、ワークショップを開催することとなりました。